



SEMESTER END EXAMINATIONS - MARCH 2022

Program	: B.E. : Computer Science and Engineering	Semester	: V
Course Name	: Database Systems	Max. Marks	: 100
Course Code	: CS52	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.
- Write figures wherever necessary.

UNIT- I

- Explain the types of attributes with suitable examples. CO1 (05)
 - Describe the three-schema architecture with a neat diagram. CO1 (08)
 - Explain why mapping is required between schema levels.
 - Draw ER Diagram for the following Statements with Structural constraints. CO1 (07)
 - A man can be married to at most one women and vice versa.
 - An airport lies in exactly one country. A country may have arbitrarily many airports (and maybe none at all)
 - One instructor teaches many courses, but each course is run by exactly one instructor.
 - Each department has exactly one department head, some employees are the head of one department.
- Draw the Entity- Relationship Diagram (ERD) for the following scenario: CO1 (08)
 - A salesperson may manage many other salespeople. A salesperson is managed by only one salespeople. A salesperson can be an agent for many customers.
 - A customer is managed by one salespeople. A customer can place many orders. An order can be placed by one customer. An order lists many inventory items. An inventory item may be listed on many orders.
 - An inventory item is assembled from many parts. A part may be assembled into many inventory items. Many employees assemble an inventory item from many parts. A supplier supplies many parts. A part may be supplied by many suppliers.
 - Discuss the main characteristics of the database approach. CO1 (06)
 - Classify any three Actors on the scene. Briefly explain each with their responsibilities. CO1 (06)

UNIT - II

- Explain Division operation of Relational algebra with an example. CO2 (05)
 - Define entity integrity and referential integrity constraints. Why are they important? CO2 (06)
 - Consider the following publishing schema CO2 (09)
 - author (authorId; first name; last name)
 - author_pub(authorId; pubId; authorPosition)
 - book (bookId; book title; month; year; editor)
 - pub (pubId; pubName; bookId)

Write relational algebra queries for the following.

- i) The names of all authors who are not book editors.
- ii) The names of all authors who have at least one publication in the database.
- iii) The titles of books published by Macmillan.

4. a) What is Union compatibility? Explain set difference operation in relational algebra with an example. CO2 (05)
- b) Write a note on Aggregate functions and Grouping in Relational algebra. CO2 (05)
- c) Consider the schema given below: CO2 (10)
 - Student (USN, Name, Branch, CGPA)
 - Faculty (F_ID, Name, Department, Designation, salary)
 - Course (C_ID, Name, F_ID)
 - Enroll (C_ID, USN, Grade)

Give the relation algebra expression for the following:

 - i. List the USN and names of students who have secured 'B' grade in CS513 course.
 - ii. List the names and salaries of faculties who are professors.
 - iii. For each department, find the maximum salary of instructors in that department.
 - iv. List the courses and the number of students enrolled for that course.

UNIT - III

5. a) Consider the schema given below for a aircraft database. CO3 (08)
 - Aircraft (aircraft ID, aircraft name, cruising range)
 - Certified (emp ID, aircraft ID)
 - Employee (Emp ID, ename, salary)

Write the SQL statements for the following queries:

 - i. Find the employee ID's of employee who make the highest salary.
 - ii. Find the employees who are certified for the largest no of aircrafts.
 - iii. Arrange the flight no with respect to the ascending order of distance.
 - iv. Find the name of pilots who can operate flights with a range greater than 3000 miles but are not certified on any Boeing aircraft.
- b) Explain the need for assertions. Create an assertion in SQL for the following statement and explain. CO3 (06)

"The salary of an employee must not be greater than the salary of the manager of the department that the employee works for"
- c) Explain any three clauses of SELECT FROM WHERE in SQL with an example. CO3 (06)
6. a) Explain the purpose and syntax of the following commands in SQL by considering an relational schema that has at least one numerical attribute. CO3 (10)
 - i)Delete ii)Update iii)Alter iv) INSERT.

2.6)

Consider the following tables

CO3 (10)

BRANCH(branch-name:string, branch-city:string, assets:real)

ACCOUNT(accno:int, branch-name:string, balance:real)

DEPOSITOR(customer-name:string, accno:int)

CUSTOMER(customer-name:string, customer-street:string, customer-city:string)

LOAN(loan-number:int, branch-name:string, amount:real)

BORROWER(customer-name:string, loan-number:int)

Write the SQL for the following

- List in alphabetical order all customers who have a loan at Perryridge branch.
- Find the names of all customers who street address includes the substring "Main".
- List the loan numbers in descending order of amount.
- Find the average account balance at each branch.
- Find the name of the customer who has the highest balance in the bank.

UNIT - IV

- What is the need for normalization? Explain 1NF, 2NF, 3NF with examples. CO4 (10)
 - Consider the universal relation $R = \{A, B, C, D, E, F, G, H, I, J\}$ and the set of functional dependencies $F = \{ \{A, B\} \rightarrow \{C\}, \{A\} \rightarrow \{D, E\}, \{B\} \rightarrow \{F\}, \{F\} \rightarrow \{G, H\}, \{D\} \rightarrow \{I, J\} \}$. What is the key for R? Decompose R into 2 NF and then 3 NF relations. CO4 (10)
- State the informal guidelines for relational schema design. Illustrate how violation of these guidelines may be harmful. CO4 (10)
 - Explain Multi-valued Dependencies, Fourth Normal Form, Join Dependencies and Fifth Normal Form with examples. CO4 (10)

UNIT - V

- What is meant by the concurrent execution of database transactions in a multiuser system? Discuss why concurrency control is needed, and give informal examples. CO5 (10)
 - Which of the following schedules is (conflict) serializable? For each serializable schedule, determine the equivalent serial schedules. CO5 (10)
 - $r_1(X); r_3(X); w_1(X); r_2(X); w_3(X);$
 - $r_1(X); r_3(X); w_3(X); w_1(X); r_2(X);$
 - $r_3(X); r_2(X); w_3(X); r_1(X); w_1(X);$
 - $r_3(X); r_2(X); r_1(X); w_3(X); w_1(X);$
- Write a note on binary locks and explain with an example. CO5 (06)
 - Discuss the actions taken by the read_item and write_item operations on a database. CO5 (06)
 - What is the system log used for? What are the typical kinds of records in a system log? What are transaction commit points, and why are they important? CO5 (08)

$A \rightarrow DE$

$B \rightarrow$